



Scan Heads



Installation and Operation Manual





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Important Information and Delivery

Product Overview

Congratulations and thank you for purchasing a General Scanning Solutions Scan Head. The configuration you have received can be identified by the label found on the side of the product. The label identifies the model name, part number, mirror coating and serial number.

About This Operating Manual

Please read these instructions carefully before installing and operating the scan head. If there are any questions regarding the contents of this manual, please contact Cambridge Technology Inc. Keep the manual available for servicing, repairs and product disposal. This manual should accompany the product if ownership changes hands. Cambridge Technology Inc. reserves the right to update this operating manual at any time without prior notification.

ESD Warning



Many of the OEM electronics manufactured by Cambridge Technology Inc. including galvanometers and servo controllers are electrostatic discharge (ESD) sensitive. Please handle these electronics with care as improper handling could cause damage to these electronics. Cambridge Technology Inc. has implemented procedures and precautions for handling these devices and we encourage our customers to do the same. Upon receiving your components, you should note if it is packaged in an ESD-protected container with the appropriate ESD warning labels. The equipment should remain sealed until the user is located at a proper static control station*. Also, please note that equipment returned to the factory must be shipped in anti-static packaging.

(*) A proper static control station should include:

- A soft grounded conductive tabletop or grounded conductive mat on the tabletop.
- A grounded wrist strap with the appropriate (1 Meg) series resistor connected to the tabletop mat and ground.
- An adequate earth ground connection such as a water pipe or AC ground.
- Conductive bags, trays, totes, racks or other containers used for storage.
- Properly grounded power tools.
- Personnel handling ESD items should wear ESD protective garments and ground straps.

Unpacking

The package you received includes those items listed on the packing list. After receiving this product, please be sure to:

- Carefully unpack the contents from the box.
- Save shipping container and packaging material in case you need to return unit for service.
- Protect the scan head from dust and other contaminants.

- Check contents of the box against the packing list to ensure all parts were received.
- Inspect each item to ensure it was not damaged during shipment.

Warranty Information

The Customer shall examine each shipment within 10 days of receipt and inform Cambridge Technology Inc. of any shortage or damage. If no discrepancies are reported, Cambridge Technology Inc. shall assume the shipment was delivered complete and defect free. Cambridge Technology Inc. warrants products against defects up to 1 year from manufacture date, barring unauthorized modifications or misuse. Repaired product is warranted 90 days after the repair is made, or one year after manufacture date - whichever is longer. Please contact Customer Service to obtain a Return Materials Authorization (RMA) number before returning any product for repair. All orders are subject to the Cambridge Technology Inc. Terms and Conditions and Limited Warranty. Please visit our website for the latest version of these documents and other useful information.

www.camtech.com

IMPORTANT: Customers assume all responsibility for maintaining a laser-safe working environment. OEM customers must assume all responsibility for CDRH (Center for Devices and Radiological Health) certification.

Customer Support

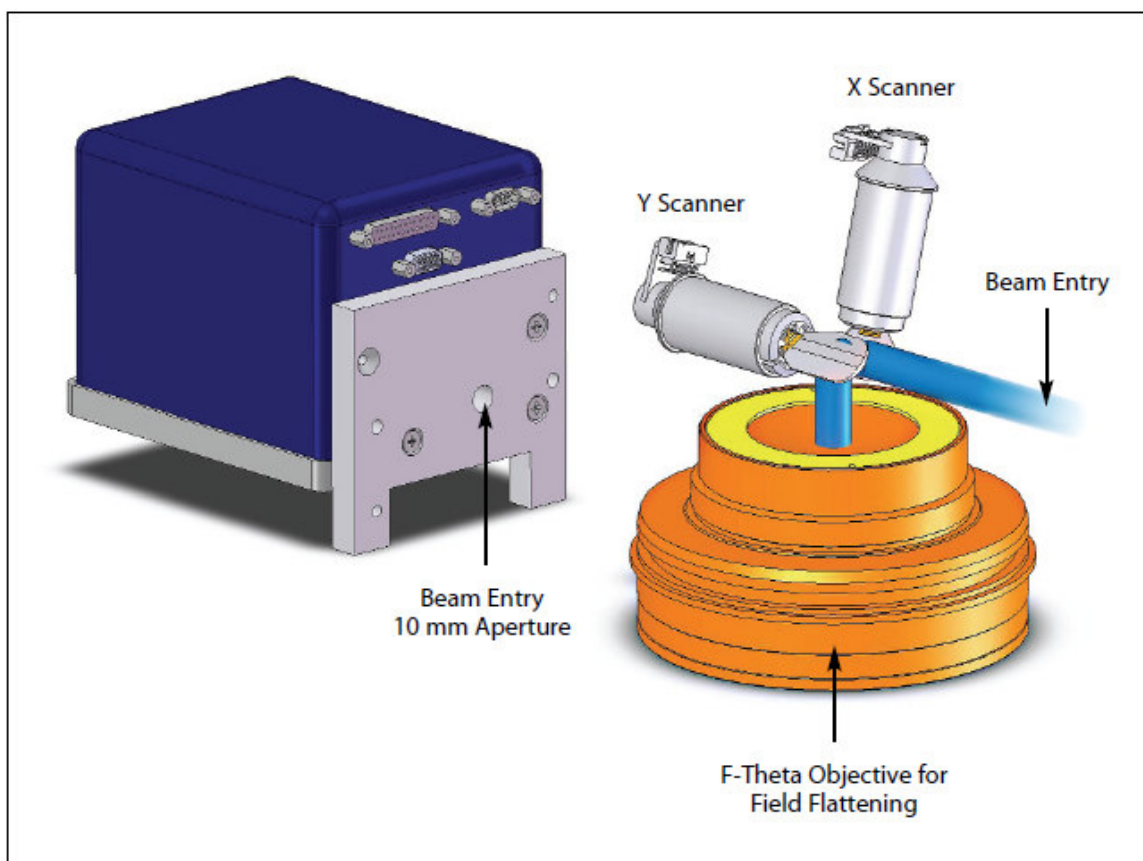
Cambridge Technology Inc. has support services to address your questions or concerns with either the product or manual you are using. Before calling for assistance, be sure to refer to any appropriate sections in the manual that may answer your questions. Please call one of our three worldwide offices between 9am and 5pm local time.

Contact Information

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		ScannerSales-Asia@gsig.com

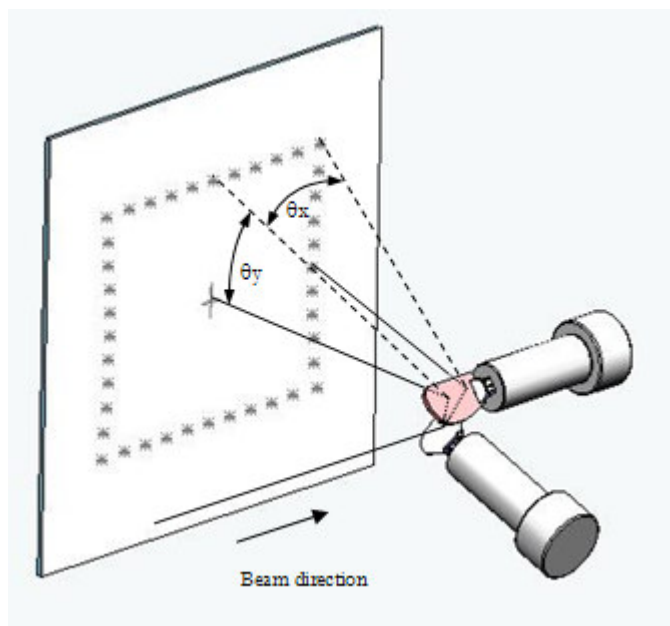
Introduction and Product Description

General Scanning Solutions Scan Heads provide the capability of deflecting optical beams in a XY plane for all possible laser applications. The laser beams are typically focused with a lens onto a working plane or planes. The synchronized actions of two galvanometer servo-controlled mirrors direct the laser beam to specific locations on a target in both the X and Y directions. At its core, a scan head contains two galvanometer scanners (X and Y) and a servo driver for each axis. There are two types of control interfaces to most scan heads: Digital XY2-100 communication and Analog Voltage differentials, typically $\pm 10V$ input sensitivity.



Principles of Operation

The beam deflection is caused by the mirrors attached to the two galvanometers. The beam enters the scan head through the input aperture and is first deflected in the X direction and then in the Y direction.



Uncorrected, the scan head will produce a projected grid that is distorted proportional to the deflection angle. This is due to the finite spacing of the galvanometer mirrors and the optical distortion induced by the F-Theta lens. Vector command controllers such as the EC1000 from Cambridge Technology Inc. use correction tables to pre-process the command data to remove these distortions. Cambridge Technology Inc. can provide correction tables for common scan head and lens configurations that are compatible with both our controllers and other commercial controllers. Also, we can create tables for custom configurations on request.

Scan Head Installation and Operation

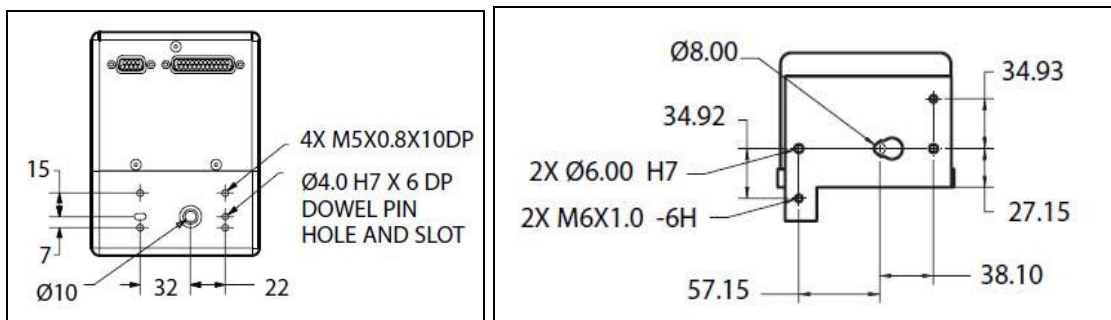
System Overview

A basic galvo based laser system is comprised of the following fundamental components. This manual contains information on the connections with the scan head.

- Laser
- Laser Power Supply
- Beam Expander (depending on laser)
- Scan Head
- F-theta Focusing Lens + Lens Spacer
- Scan Head Power Supply
- System Controller – sends synchronized signals to the laser and the scan head
- Grid Calibration File
- PC or Laptop – To control/program System Controller
- Cables
 - Controller → Laser
 - Controller → Scan Head
 - Laser Power Supply → Laser
 - Scan Head Power Supply → Scan Head

Mechanical

General Scanning Solutions Scan Heads have two metric dowel pin holes for alignment in addition to two or four tapped metric holes for mounting. There are two standard mechanical laser interface patterns used on General Scanning Solutions Scan Heads: Traditional and Compact. Some scan heads can be optioned with either Traditional or Compact laser plates while others come with only one laser interface. Adapter plates are available to convert from one pattern to another. We recommend creating a mounting flange that will match the mounting holes on the beam input side of the assembly. Use the model specific outline drawing or the 3D Customer CAD model to determine the dimensions for the manufacture of a mounting flange.

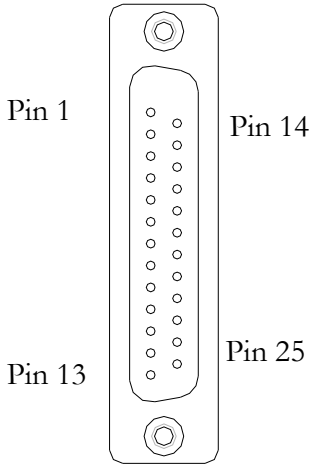


Electrical – Communication

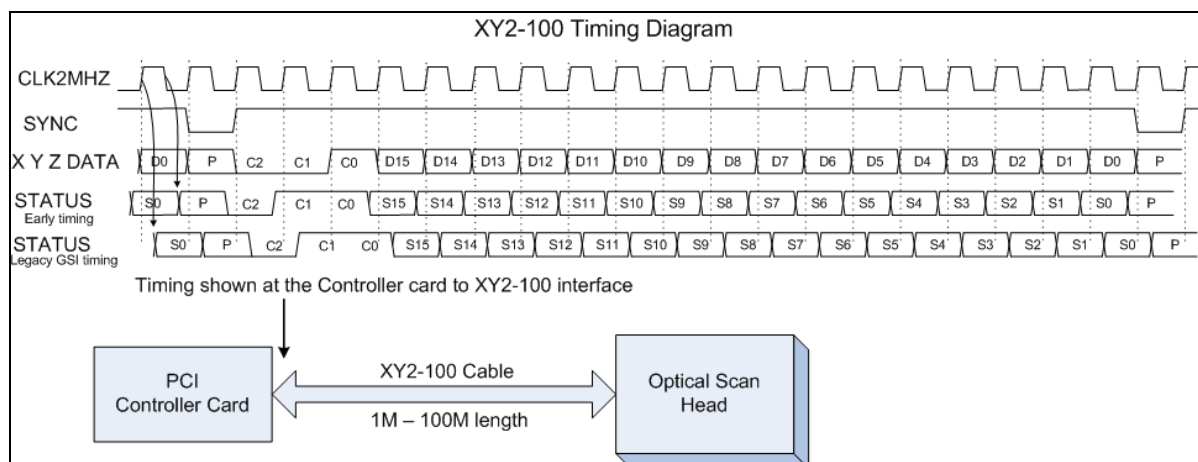
General Scanning Solutions Scan Heads are available with two types of control interfaces to most scan heads: Digital XY2-100 Communication and Analog Voltage differentials, typically +/-10V input sensitivity.

Digital XY2-100 Communication

Digital XY2-100 Communication is available via a female 25pin Dsub connector. The scan head has the following pin assignment.

Interface	Pin	Assignment
 Pin 1 Pin 14 Pin 13 Pin 25 25-Pin D-sub (Female) Digital Connection	1	<u>XY2-100 Connections</u> CLOCK -
	14	CLOCK +
	2	SYNC -
	15	SYNC +
	3	DATA X -
	16	DATA X +
	4	DATA Y -
	17	DATA Y +
	5	DATA Z -
	18	DATA Z +
	6	STATUS -
	19	STATUS +
		<u>Reserved Pins</u>
	7-13	N/A
	20-25	N/A

The XY2-100 compatible serial protocol uses a clock line CLOCK, a frame SYNC signal, three data channels CH1 (Data-X), CH2 (Data-Y) and optional CH3 (Data-Z) and a STATUS channel. (Note: Status data type and availability is model specific.) The vector controller used to drive the scan head will deliver set values for X, Y and the optional Z axis. The timing diagram below illustrates the relationships for the XY2-100 signals.



CLOCK - The Clock is transmitted by the position data generator, 20 cycles per frame. The nominal frequency is 2MHz.

SYNC - The frame Sync is a single logical "0" pulse, once per frame, transmitted by the position data generator one clock cycle prior to the first bit of the frame.

X Y Z DATA - The X, Y, & Z Data are three 20-bit serial data streams consisting of a 3-bit control code, one 16-bit position word (unsigned, MSB first), and a parity bit (even parity).

Improper control codes, parity error, or a missing sync on the 21st clock cycle will cause the position data word for the affected channel to be discarded. 001 is the only supported control code as shown in the following table.

Control Bits [2:0]	Function
0 0 0	Reserved
0 0 1	16-bit position data follows
0 1 0	Reserved
0 1 1	test (reserved)
1 x x	Reserved (extended addressing)

Scan Head Status Signal

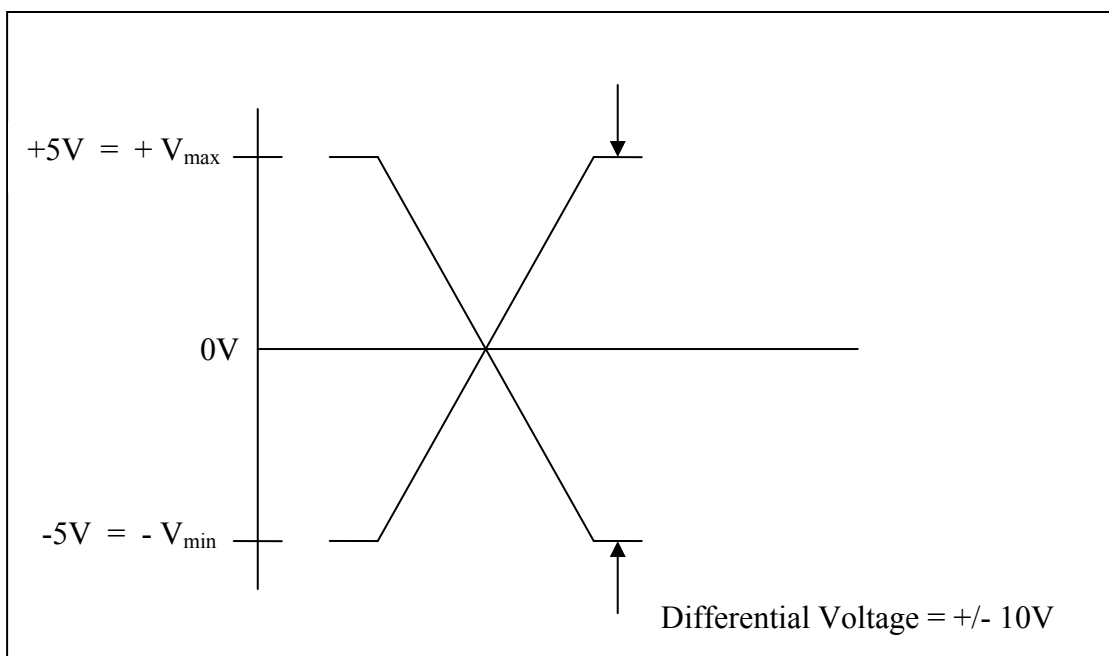
The status channel provides several error messages indicating scanner or servo exceptions. When an error condition occurs, the corresponding bit is set to low level. When the error is cleared, the corresponding bit is set to high level. After power up or reset the status line is held low until the Interface is up and running. Table 1 defines the meaning of the status information. Status signals are scan head specific.

Table 1 – Status Bit Definitions

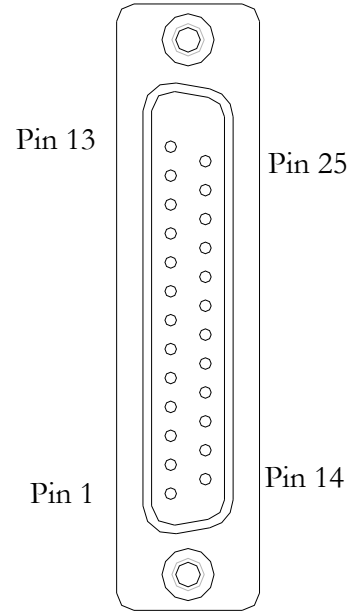
Status Bit	Definition
PwrErr	Indicates if SmartDirect interface is up and running.
TmpErr	Galvo coil temperature error. This error indicates that the motor coil temperature is close to maximum. The system will proceed until the critical temperature is reached and then it will shut down.
X/Y-PosErr	Position error. Occurs whenever the motor position differs from the command by a certain limit. This limit is a system parameter.

Analog Voltage Communication

Analog Voltage Differential Communication is available via a male 25pin Dsub connector on most scan head models. The analog command configuration requires a differential input signal of $\pm 10V$ which equals the scale of the system. The differential input signal can be understood in the figure below.



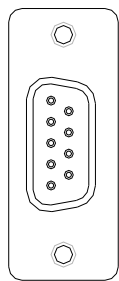
Analog Voltage Communication is available via a male 25pin Dsub connector with the following pin assignment.

Interface	Pin	Assignment
 25-Pin D-sub (Male) Analog Connection		<u>X Axis Connections</u>
	13	+X COMMAND
	25	-X COMMAND
	11	SIGNAL GROUND
	20	SERVO ENABLE
	23	SERVO READY
	9	+X SIGNAL READBACK*
	7	-X SIGNAL READBACK*
		<u>Y Axis Connections</u>
	12	+Y COMMAND
	24	-Y COMMAND
	10	SIGNAL GROUND
	21	SERVO ENABLE
	22	SERVO READY
	8	+Y SIGNAL READBACK*
	6	-Y SIGNAL READBACK*
		<u>Reserved Pins</u>
	1 - 5	N/A
	14 - 19	N/A
		*Signal type and availability is model specific.

Command Input - The input impedance looking into one of the command input pins while having the opposite grounded is scan head model specific.

Electrical – Power

Power is supplied to the scan head via a Male 9pin Dsub connector. The power supply should be chosen to match the specifications of each scan head. Some scan heads can accept a symmetrical supply of +/-24V whereas typically they are tuned for +/-15V. It is important to use the correct voltage for the scan head. Using a higher voltage than specified requires either a different servo tuning or additional heatsinking. Additionally, it may lead to faulty behavior in the system.

Interface	Pin	Assignment
 9-Pin D-sub (Male) Power Connection	4, 5, 9	+Vin
	3, 7, 8	GND
	1, 2, 6	-Vin

Electrical – Power Supply Requirements

The maximum RMS current required is approximately 3Amps, but peak current demands can approach 10Amps on certain models.

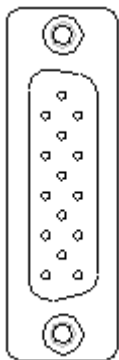
Electrical – Tuning and Service Ports

General Scanning Solutions Line of Lightning Digital Scan Heads have a third port for remote access to the digital servo.

Lightning Digital Scan Heads TuneMaster Port

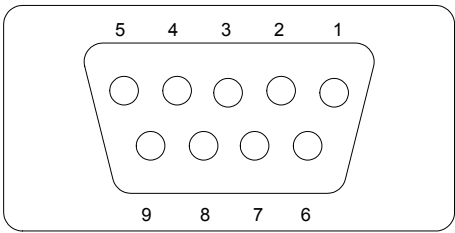
The TuneMaster port is accessible by a Male High Density 15pin Dsub connector. Please see the TuneMaster manual for directions. The pin assignment is shown below.

Additionally, a TuneMaster cable can be purchased (part number 712-78977) that connects a PC or laptop via 9pin serial port to the 15pin TuneMaster port.

Interface	Pin	Assignment
 15-Pin HDB (Male) Communication & Probes Connection	1	<u>Connections</u> [Reserved]
	6	[Reserved]
	11	D-GND 2
	2	TXD
	7	-CTS
	12	[Reserved]
	3	RXD
	8	-RTS
	13	[Reserved]
	4	Probe 1
	9	Probe 2
	14	D-GND 1
	5	Y Position
	10	A-GND
	15	X Position

LightningXP Digital Scan Heads Service Port

The LightningXP Service port is accessible by a Female 9pin Dsub connector. The service port is used for two key functions: remote access and laser delay. Please see the DC2000 user manual for more detailed information. The pin assignment is shown below.

Interface	Pin	Assignment
 9-Pin (Female) Service Port		<u>Connections</u>
	1	LD1-IN
	2	RXD
	3	TXD
	4	LD2-IN
	5	DGND
	6	LD1-OUT
	7	J1A
	8	J1B
	9	LD2-OUT

Service Port For Laser Delay

The LightningXP Digital Scan Head uses the DC2000 self-tuning State-Space digital servo to maximize the performance of the galvanometers. The DC2000 servo uses a zero tracking error approach to galvo movement by estimating the galvo motion based on a model of the system and calculating the right voltages to reproduce the commanded input. When the commanded input requires a galvo voltage greater than the galvo power supplies the DC2000 uses different filters to drive the galvo within the limits of the available power supplies and enable maximum speed. The filters operate by placing the galvo command stream into a pipeline buffer where the input sequence of motion commands are analyzed together, system parameters are adjusted in real-time to maximize performance for the system galvo, mirrors and power supplies, and then the optimized command output is driven out to the galvanometers. Because this pipeline delay is a constant 380 μ s, this “pipeline delay” between the command signal and the galvo movement does not alter galvo motion, mark timing or character formation. It merely shifts the marking session forward by 380 μ s. To keep the laser control and galvo motion in sync, a 380 μ s must be applied to the laser signals as well. The 9pin Service Port can be used to delay the laser commands by 380 μ s by passing the laser signal lines through the scan head. There are 2 sets of “laser delay” 3.3v logic lines (LD1-in/out and LD2-in/out) supported by the 9 Pin Service Port. Alternatively, a 380 μ s delay to the laser lines may be programmed into your controller directly.

Optical – Laser Beam Diameter

General Scanning Solutions Scan Heads have a specified Clear Aperture. The clear aperture is the maximum beam diameter that can enter the scan head. This clear aperture is the size of the opening before the beam reflects off the first mirror. The mirrors are designed to reflect the entire beam off the X mirror and the Y mirror and out the bottom (minus losses due to mirror reflectivity). There is an additional tolerance added around

the edge of the mirror to ensure that the beam will not fall off the mirror surface at any point within the specified scanning angle for that scan head.

Optical – System Scaling

General Scanning Solutions Scan Heads are optimally designed for both high speed and large field sizes. Each scan head model has mirror designs to achieve these goals. The following table shows the optical mirror scanning capability in degrees by clear aperture size.

Clear Aperture	7mm	10mm	14mm	20mm	25mm
System Scaling	+/-22°	+/-22°	+/-22°	+/-20°	+/-15°

Each of the systems with the Digital XY2-100 Communication are optically scaled to a full 65,535 bits and the Analog Voltage Communication systems are optically scaled to a full +/-10V.

For example, a Digital XY2-100 Communication system with +/-22° mirrors results in a system resolution of...

$$\begin{aligned}
 \pm 22^\circ \text{ optical} &= \pm 32,767 \text{ bits} \\
 44^\circ \text{ optical} &= 65,535 \text{ bits} \\
 1^\circ \text{ optical} &= 1,489.4 \text{ bits}
 \end{aligned}$$

An Analog Voltage Communication system with +/-22° mirrors results in a system resolution of...

$$\begin{aligned}
 \pm 22^\circ \text{ optical} &= \pm 10\text{V} \\
 44^\circ \text{ optical} &= 20\text{V} \\
 1^\circ \text{ optical} &= 0.4545 \text{ V}
 \end{aligned}$$

For example, a Digital XY2-100 Communication system with +/-15° mirrors results in a system resolution of...

$$\begin{aligned}
 \pm 15^\circ \text{ optical} &= \pm 32,767 \text{ bits} \\
 30^\circ \text{ optical} &= 65,535 \text{ bits} \\
 1^\circ \text{ optical} &= 2,184.5 \text{ bits}
 \end{aligned}$$

An Analog Voltage Communication system with +/-15° mirrors results in a system resolution of...

$$\begin{aligned}
 \pm 15^\circ \text{ optical} &= \pm 10\text{V} \\
 30^\circ \text{ optical} &= 20\text{V} \\
 1^\circ \text{ optical} &= 0.6667 \text{ V}
 \end{aligned}$$

This distinction is important to note. If you change from a 14mm to a 20mm system, a 1000 bit command for example will yield a shorter line in the 20mm system than in the 14mm system.

Optical – Mirror Cleaning



Although the user can replace the mirrors, we do not recommend you do so. Furthermore, Cambridge Technology does not recommend cleaning front surface mirrors. Mirrors damaged by cleaning are not included under the warranty. The surface of these mirrors damage easily. Prevention of hard dust particles from being entrained in the process and causing scratches is difficult. In many cases, small defects in the mirror's surface may be less harmful than the surface damage resulting from continued cleaning. Cleaning requires special equipment typically not readily available to most users. However, there are times when cleaning the mirror becomes a necessity, e.g. stains such as fingerprints must be removed

immediately to prevent permanent etching of the reflective surface. The information below includes general recommendations for those special occasions when mirrors must be cleaned. Remove lint from mirrors with a jet of low pressure clean air or nitrogen. Take caution not to blow on the front surface of mirrors with your mouth as this can deposit moisture that may stain the finish. A thin overcoating of silicon monoxide protects most mirrors from oxidation. Like many optical coatings, it is easily damaged when attempts are made to clean the mirror surface with a dry tissue. The safest method of cleaning is to place a piece of lens tissue wet with reagent grade (highly pure) alcohol. Lay the wet tissue over the surface of the mirror, gently agitate it, then slide the tissue off. If an uneven film remains when dry, repeat the process, but use a jet of low pressure clean air or nitrogen to quickly spread remaining liquid. This should remove the problem blemishes.

- Note that the mirror is not rubbed.
- Do not let solvent enter the bonded zone of the mirror.
- Do not let solvent enter the scanner bearings. When wetting the mirror's surface, hold the Scan Head at an angle so that the liquid does not wet the scanner. Solvent found in the bearings of the scanner voids the warranty.

Optical – Mirror Coatings

Cambridge Technology Inc. offers a wide variety of laser quality first surface mirrors for multi-axis applications. All Cambridge Technology Inc. mirrors are mounted in a mass balanced configuration for optimum scanning performance. The following standard coatings are available. Other coatings are available on request.

S4: Durable Silver 2 Coating

YH: Nd:Yag High Power Coating with Visible Alignment Band

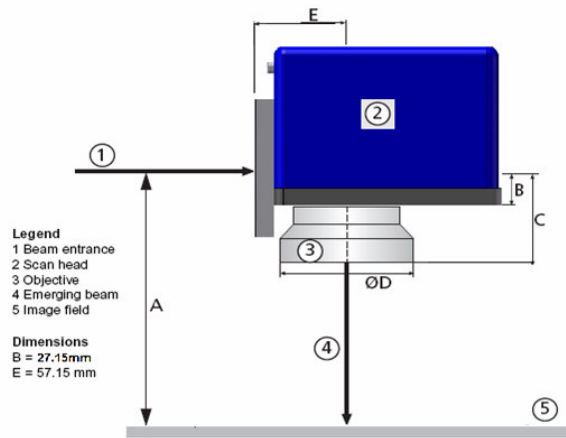
TH: Nd:Tripled Yag Coating with Visible Alignment Band

Optical – Lens Options

General Scanning Solutions family of scan heads are available with a wide range of standard f-theta lenses and lens spacers. The positioning of the lens relative to the scan

head is critical for both scanning performance and safety of the scanning mirrors. Our lenses and lens spacers are optimized for field size while ensuring there are no major or minor back reflections from the optics onto the surface of the mirrors which can cause damage. General Scanning Solutions has a wide range of standard lens spacers and we also create custom spacers for non-standard lenses. Our lens selection sheet for each scan head model has the following information.

- Working distance from bottom of the scan head to the working surface
- Maximum angular range in optical degrees through the lens
- Maximum square scan field (mm^2)
- Maximum diagonal field (mm)



If you would like to create your own spacer, please refer to the outline and installation drawing of the scan head or the Customer 3D CAD model to position the lens in a safe location. You will need the following information about the scan head to position the lens.

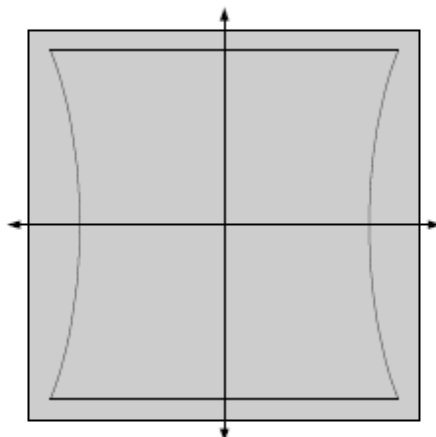
- Distance between center of X and center of Y mirror at nominal position
- Distance between center of Y mirror and bottom surface of the scan head
- Distance from bottom of the scan head to first mechanical interference (no lens zone)

Optical – Grid Correction

Grid correction is needed to compensate for optical errors induced by all two axis laser beam systems. These optical distortions are caused by a number of factors, including the distance between each mirror, the distance between the mirrors and the image field, and the type of lens used in the laser for focusing the laser beam.

Pincushion Distortion

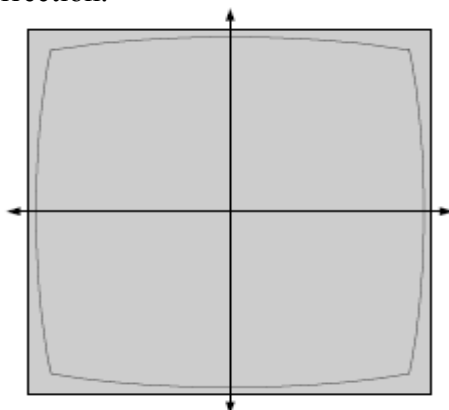
The distortion caused by a 2 axis X-Y galvo system is known as pincushion distortion. The distortion in the X axis is proportional to the tangent of the angle of the Y axis mirror and the distance from Y mirror to the focal plane. This distortion can be compensated for by grid correction.



Pincushion distortion caused by X-Y mirror set

F-theta Objective Induced Distortion

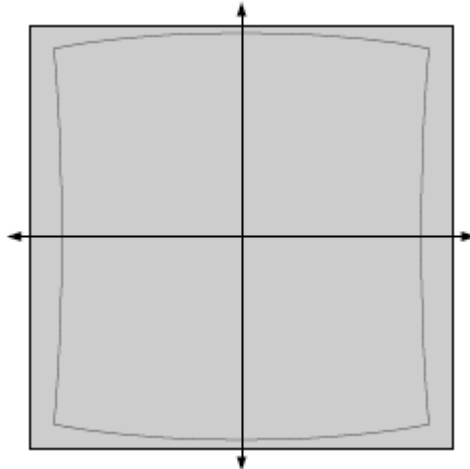
The other major type of distortion is caused by the addition of an f-theta objective to an X-Y galvo system. F-theta objective lenses, like all optical lenses, are not perfect and induce their own projection field distortions. This distortion, illustrated in Figure 8, is called “pillow” distortion for what it does to a square image. In reality, this distortion is radially symmetric from the image field origin and can often be modeled as a third order polynomial. Many projection lens vendors will provide these model coefficients, or measurement data from which these coefficients can be derived. This distortion can be compensated for by grid correction.



Pillow distortion caused by F-theta lens

Grid Correction – Composite Distortion and Correction Methodology

The two components of distortion described above combine to create an image field similar to that shown below. This distortion can be compensated for through the use of grid correction tables.



Composite Image Field Distortion

Correction tables represent a 65x65 element grid covering the full addressable projection range of the system. Each grid element contains three correction components: one each for the X, Y and Z axes. The components represent an offset that if added to an ideal position command for that point, would alter the galvo positions such that the resulting projected point would fall onto a “perfect” grid, i.e. the point would be “corrected.”

Correction Tables

General Scanning Solutions provides grid correction tables for common scan head and lens configurations that are compatible with both CTI controllers and other commercial controllers. Also, we can produce tables for custom configurations on request. We can supply a default correction table to cover the mirrors and lens distortions in either a Comma Separated Values Format (CSV) or XML format for the X, Y and Z axis distortion.

Correction Table Examples:

CSV format: X-axis, Y-axis, Z-axis
 1763, 0, 0
 1708, 0, 0
 1653, 0, 0
 1598, 0, 0

XML format: <Data type="CorrTableData" rev="1.0">
 <x-axis>1763,1708,1653,1598</x-axis>
 <y-axis>0,0,0</y-axis>
 <z-axis>0,0,0</z-axis>
 </Data>

Safety

Safety During Installation

To reduce the risk of injury, please observe the laser safety guidelines suggested in this section. Where possible, the General Scanning Solutions Scan Head should be inter connected for software shut down of the laser. This will ensure safety during operation, error or recovery of the scan head. In all cases, we recommend that you fully enclose and interlock the zone of hazard for your application to prevent possible beam deflections while the laser is energized. Refer to ANSI Z136.1 to determine what protective equipment is required. At no time should you stare into the beam, place any parts of your body in the beam path, or expose yourself to reflections of powerful beams. You should use a Class 1 HeNe Laser for alignment. If this is not possible, you should use the available laser's lowest power. Using optical instruments with this product increases eye hazard. Additional Safety requirements may be applicable during initial alignment of the optical system. Final analysis of the system should be performed by a Laser Safety Officer, or a competent specialist in this field.

Operational Guidelines and Standards

When operating the scan head, the following guidelines and standards apply:

- EC-Guideline 73/23/EWG
- Low Voltage Guidelines
- EC-Guideline 89/336/EWG
- Electromagnetic Compatibility (including revision 92/31/EWG)
- EC-Guideline 89/392/EWG
- Machine Guidelines (including revisions 91/368/EWG and 93/44/EWG)
- EN 60825 Laser Equipment Safety, Part 1 (1994): Classification of Equipment, Requirements and User Guidelines (VDE 0836)
- EN 60204 Electrical Equipment Part 1: General Requirements (June 1993)
- Complying with the Relevant Standards for the CE Label

In addition, A technical survey of laser safety requirements can be found in ANSI Z136.1, "American National Standard For the Safe Use of Lasers." This is available from:

American National Standards Institute, Inc.
1430 Broadway
New York, New York 10018
www.ansi.org

Among the many other sources of laser safety information, the following institution offers several excellent publications:

The Laser Institute of America
5151 Monroe Street, Suite 118W
Toledo, Ohio 43623
www.laserinstitute.org

Please note: The scan head is delivered as an OEM component for integration into a laser scanning system. The system manufacturer bears the responsibility for complying with the standards and guidelines required for the CE label. Please contact Cambridge Technology Inc. for further information about this product and applicable guidelines.

Laser Safety

This scan head is designed to be operated in conjunction with a laser. Therefore, all applicable rules and regulations for safe operation of lasers must be known and applied when installing the scan head and operating the system in which it is used. Since Cambridge Technology Inc. has no influence over the employed laser or the overall system, the customer is solely responsible for the laser safety of the entire system.

Laser Shutter

The scan head has no shutter and there is no device to decrease the laser output power. Because each laser is unique, it is the responsibility of the user to include such a device as required. CFR 1040.10 [f] [6] states: “A beam attenuator is required on Class II, IIIa, IIIb and IV laser systems. The beam attenuator is a mechanical or electrical device such as a shutter or attenuator that blocks emission. The beam attenuator blocks bodily access to laser radiation above Class I limits without the need to turn off the laser. The beam attenuator must be available for use at all times during operation. Power switches and key controls do not satisfy the attenuator requirement.” Please refer to CFR 1040.10 for additional information.

The beam shutter should be installed between the laser and the Scan Head. The figure below shows the recommended location of the shutter. We strongly recommend that you specify a laser with a vendor-supplied shutter mechanism. If this is not possible, consult the laser vendor to design a proper safety shutter.

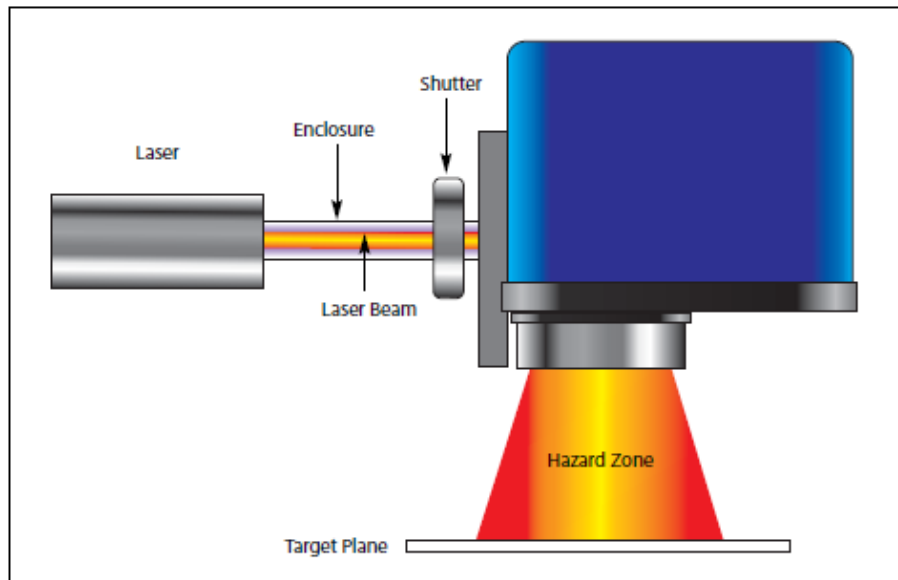


Figure: The laser's internal and external optical path towards the target plane, specifically where the hazard zones are located as the optical beam passes through the scan head.